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## System Integration and Architecture Project Proposal

### Development of a Data-Driven Decision Support System for Laguna Livestock Farmers

#### Introduction

This project aligns with SDG 2: Zero Hunger by improving food security through data-driven insights into livestock production, market demands, and risk mitigation (e.g., bird flu seasons). It gives smallholder farmers the ability to maximize livestock health, pricing, and supply timing for sustainable agriculture.

The initiative supports farmers in Laguna province who raise poultry, produce eggs, and farm fish. Farmers will use a mobile app as the main tool, while Department of Agriculture (DA) Laguna officials will use a web app for monitoring and management. The system collects data on market prices, inflation, seasons, consumer demand, and calamities. This data is analyzed to help farmers decide the best time to sell, how much to supply, and how to avoid risks such as crop or livestock failure (“palya”). It also provides early warnings for issues like bird flu and offers guidance on improving livestock production. Farmers receive real-time alerts and recommendations through the mobile app, while DA executives access dashboards and analytics through the web app for oversight and decision-making.

#### Background

Diverse farmers in the livestock, poultry, and fisheries sectors are supported by Laguna's Department of Agriculture; however, a lack of integrated market data poses serious challenges for these producers [1], [2], [3]. In order to manage complicated risks like seasonal monsoons, changes in market demand, and disease outbreaks like Highly Pathogenic Avian Influenza (HPAI), many farmers today must rely on unofficial knowledge, which frequently leads to less than ideal profits and difficulties with their way of life [1], [3], [4]. There is a significant gap between farm-gate prices and final retail costs in certain industries, such as duck egg production, due to ineffective marketing systems that involve an excessive number of middlemen [3], [5]. Additionally, the National Capital Region (NCR) lacks a unified technology platform that would enable provincial farmers to compare prices in real time with high-demand centers [1], [6], [7].

According to research, predictive analytics can strengthen farmer resilience by combining historical price lists and seasonal risk profiles with regional inflation data, which was 3.7% for Laguna in 2023 [1], [8], [9]. Even in specialized markets with limited data, advanced Time-Series Foundation Models (TSFMs), like Time-MoE, have proven to be able to greatly

outperform conventional techniques in agricultural price forecasting [9]. Furthermore, putting in place multi-hazard early warning systems can offer “temporal explanations,” assisting farmers in determining precisely when climate hazards, such as heavy rain or heatwaves, will affect their crops or livestock [8]. The DA can close the digital divide and guarantee that local farmers are prepared for investment and able to maintain regional food security by implementing data-driven tools and programs like the KADIWA program [1], [2], [6].

## **Objectives**

The objectives of the project are to provide farmers with early information about livestock demands, price differences between Laguna province and the NCR, and bird flu seasons or prevention techniques; to forecast the best times to sell, possible shortages of supplies, chances to improve livestock growth, and the dangers of production failures; and to enable direct price comparisons between the regions to increase profits and improve farmer lifestyles.

To do this, a software solution will be created that includes a web application serving as an executive dashboard for thorough data management and visualization, and a mobile application that allows farmers to receive alerts and forecasts on the go. The system covers two levels of management: tactical, which gives Department of Agriculture (DA) executives oversight of trends for prompt interventions, and operational, which supports farmers' daily decisions about supply and sales. By facilitating planning through predictive forecasts, organizing through the integration of data on prices, inflation, market demands, and disasters, leading through early warnings and practical recommendations, and controlling through the monitoring of livestock health and market performance, it supports essential management functions.

In order to create predictive models, the data analytics component will use historical data, such as localized inflation, price lists from the NCR and province, bird flu seasonality, particular market demands for chicken and duck, and records of calamities. These models will predict possible failures, forecast supply needs, suggest livestock improvements, and provide insights on when and where to sell.

## **Scope and Limitations**

The project's scope specifically covers chicken, chicken eggs, duck, duck eggs, and fish within the Laguna region, with features for price comparisons between the province and the National Capital Region (NCR) and predictive analytics for market demands, seasonal patterns, and production risks delivered through integrated mobile and web applications.

However, it is limited by several important factors: the system will not incorporate real-time global market data; the initial development phase will primarily focus on avian livestock, with fish as a planned but subsequent expansion; and its effectiveness depends entirely on the quality and availability of the provided data.

## References

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